

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

To decide the correct resistance , use the standard expression for power dissipated in a resistor :

$P = V^2 / R .$

- Voltage : 12 V (from 6 Ã 2 V).

- Power : 40 W .

- $R = V^2 / P = (12 \text{ V})^2 / 40 \text{ W} = 144 / 40 = 3 . 6 \text{ }\Omega .$

Using Oh m âs law :

$I = V / R = 12 \text{ V} / 3 . 6 \text{ }\Omega = 3 . 333 \text{ A} \hat{=} 3 . 33 \text{ A} .$

Thus , the head light bulb draws about 3 . 33 amps and has a resistance of 3 . 6 oh ms .

Correct answer : E . 3 . 33 amps , 3 . 6 oh ms .

</think>

< answer >

E . 3 . 33 amps , 3 . 6 oh ms

</answer>

